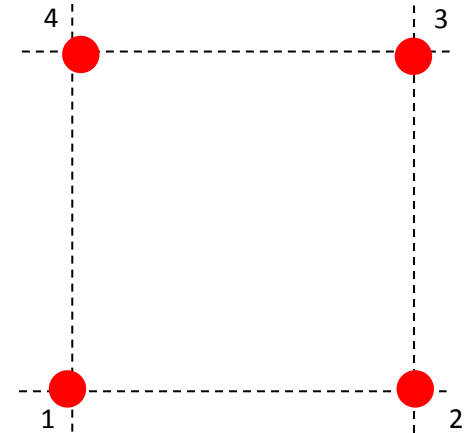


TRABAJO PRÁCTICO Nº 1

CARGA ELÉCTRICA. LEY DE COULOMB

P6. Cuatro cargas de igual magnitud $Q = 1\mu\text{C}$ están dispuestas en los vértices de un cuadrado de arista $l=0,2\text{ m}$, como se muestra en la figura.

- a) Halle la fuerza sobre la carga del vértice inferior izquierdo que ejercerán las restantes.
- b) A partir de lo encontrado en el inciso anterior, podrá responder ¿cuánto vale la fuerza que experimentan cada una de las cargas restantes debida a sus vecinas sin realizar nuevamente la cuenta?



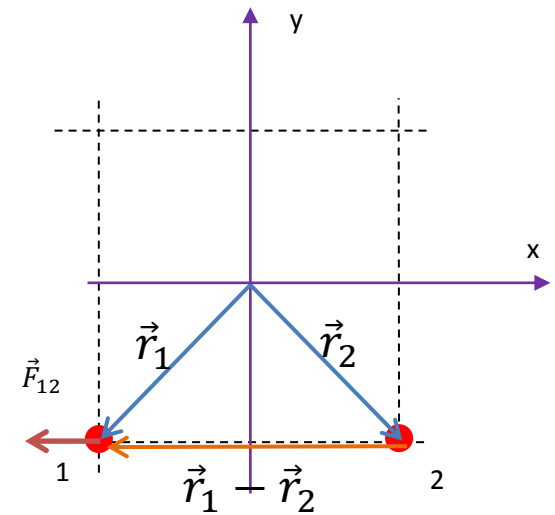
Antes de comenzar a resolver este ejercicio recordemos lo visto en la teoría para poder trabajar con la Ley de Coulomb.

$$\vec{F}_{ij} = \frac{kQ_iQ_j}{(\vec{r}_i - \vec{r}_j)^2} \frac{(\vec{r}_i - \vec{r}_j)}{|\vec{r}_i - \vec{r}_j|}$$

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14}$$

$$\vec{F}_{12} = \frac{kQ_1Q_2}{(\vec{r}_1 - \vec{r}_2)^2} \frac{(\vec{r}_1 - \vec{r}_2)}{|\vec{r}_1 - \vec{r}_2|}$$

$$\vec{r}_1 = \left(-\frac{l}{2}, -\frac{l}{2}\right) \quad \vec{r}_2 = \left(\frac{l}{2}, -\frac{l}{2}\right)$$



$$\vec{r}_1 - \vec{r}_2 = (-l, 0)$$

$$|\vec{r}_1 - \vec{r}_2| = l$$

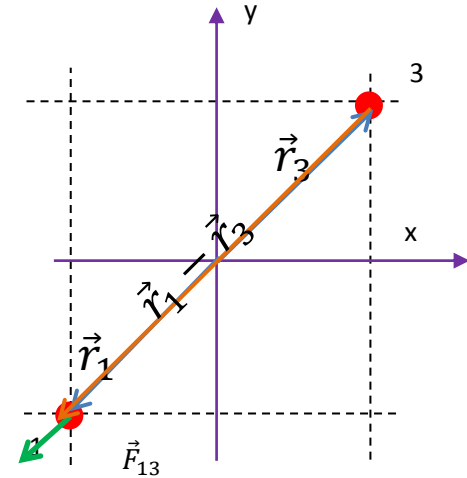
$$\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = (-1, 0)$$

$$\vec{F}_{12} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} (1\mu C)^2}{(0,2m)^2} (-1, 0)$$

$$\boxed{\vec{F}_{12} = 225mN(-1, 0)}$$

$$\vec{F}_{13} = \frac{kQ_1Q_3}{(\vec{r}_1 - \vec{r}_3)^2} \frac{(\vec{r}_1 - \vec{r}_3)}{|\vec{r}_1 - \vec{r}_3|}$$

$$\vec{r}_1 = \left(-\frac{l}{2}, -\frac{l}{2}\right) \quad \vec{r}_3 = \left(\frac{l}{2}, \frac{l}{2}\right)$$



$$\vec{r}_1 - \vec{r}_3 = (-l, -l)$$

$$|\vec{r}_1 - \vec{r}_3| = \sqrt{l^2 + l^2} = \sqrt{2}l$$

$$\frac{\vec{r}_1 - \vec{r}_3}{|\vec{r}_1 - \vec{r}_3|} = \left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

$$\vec{F}_{13} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} (1\mu C)^2}{(\sqrt{2}0,2m)^2} \left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

$$\vec{F}_{13} = 112,5mN \left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

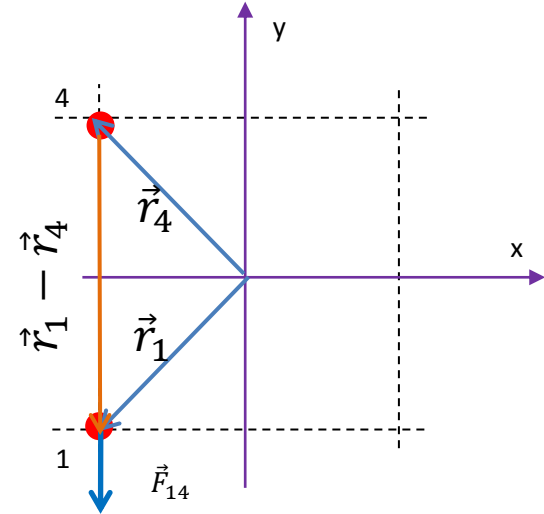
$$\vec{F}_{14} = \frac{kQ_1Q_4}{(\vec{r}_1 - \vec{r}_4)^2} \frac{(\vec{r}_1 - \vec{r}_4)}{|\vec{r}_1 - \vec{r}_4|}$$

$$\vec{r}_1 = \left(-\frac{l}{2}, -\frac{l}{2}\right) \quad \vec{r}_4 = \left(-\frac{l}{2}, \frac{l}{2}\right)$$

$$\vec{r}_1 - \vec{r}_4 = (0, -l)$$

$$|\vec{r}_1 - \vec{r}_4| = l$$

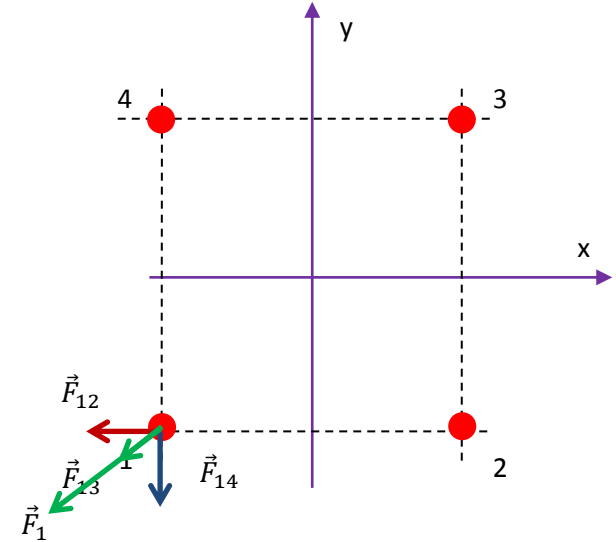
$$\vec{F}_{14} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} (1\mu C)^2}{(0,2m)^2} (0, -1)$$



$$\frac{\vec{r}_1 - \vec{r}_4}{|\vec{r}_1 - \vec{r}_4|} = (0, -1)$$

$$\vec{F}_{14} = 225mN(0, -1)$$

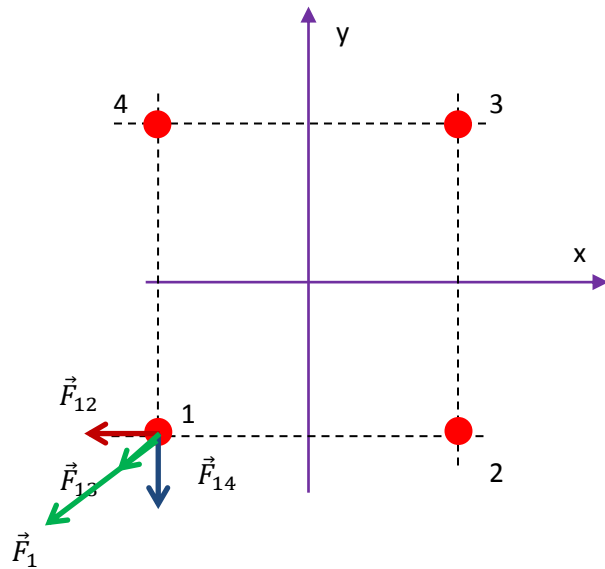
$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14}$$



$$\vec{F}_1 = 225mN(-1,0) + 112,5mN\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) + 225mN(0,-1)$$

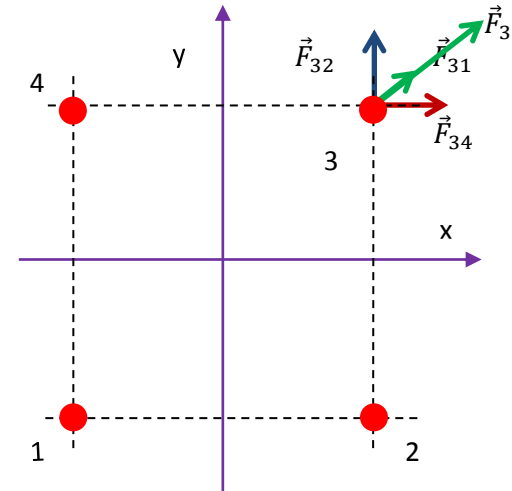
$$\vec{F}_1 = -304.5mN\vec{i} - 304.5mN\vec{j}$$

$$\vec{F}_1 = 430,7mN\left(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right)$$



$$\begin{aligned}\vec{F}_1 &= 225mN(-1,0) \\ &+ 112,5mN\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) \\ &+ 225mN(0,-1)\end{aligned}$$

$$\vec{F}_1 = 430,7mN\left(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right)$$



$$\begin{aligned}\vec{F}_3 &= 225mN(1,0) \\ &+ 112,5mN\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \\ &+ 225mN(0,1)\end{aligned}$$

$$\vec{F}_3 = 430,7mN\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$